

INTERMEDIATE EXAMINATION - 2016

(ANNUAL)

PHYSICS

I.Sc

Time- $3\frac{1}{4}$ Hours

Full Marks: 70

Instruction for the candidates:

- 1) Candidates are required to give their answers in their own words as far as practicable.
- 2) Figures in the right hand margin indicate full marks.
- 3) While answering the question, candidate should adhere to the word limit as far as practicable
- 4) 15 Minutes of extra time has been allotted for the candidates to read the questions carefully
- 5) This question paper is divided into two sections- **Section-A** and **Section-B**
- 6) In **Section-A**, there are **27 objective type questions** which are compulsory, each carrying **1 mark**. Darken the circle with blue/ black ball pen against the correct option on OMR Sheet provided to you.
Do not use Whitener/ Liquid/ Blade/Nail on OMR Paper, otherwise the result will be invalid.
- 7) In **Section- B**, there are **11 short answer type questions (each carrying 2 marks)**, out of which any **11 questions are to be answered**. Apart from this, there are **4 long Answer Type questions (Each Carrying 5 marks)**. Each question has alternate option.
- 8) Use of any electronic device is prohibited.

Section-I (Objective Type)

In the following Question no. 1 to 28 there is only one correct option on the answer sheet:
28 x 1 = 28

1. The torque acting on electric dipole moment $\vec{p}$ Placed in electric field of intensity $\vec{E}$.

(a) $\vec{p} \times \vec{E}$

(b) $\vec{p} \cdot \vec{E}$

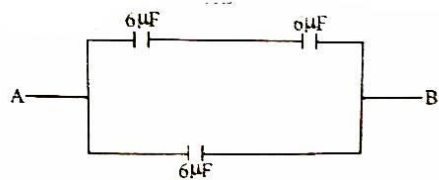
(c) pE

(d) $\vec{p} / \vec{E}$

SOL:

Correct option is A

2. The equivalent capacity between A and B in the given figure is



(a) $6\mu F$

(b) $18\mu F$

(c) $9\mu F$

(d) $\frac{1}{9}\mu F$

SOL:

Correct option is C

3. The power of electric circuit is

(a) $V.R$

(b) $V^2.R$

(c) V^2 / R

(d) $V^2.RI$

SOL:

Correct option is C

4. Dimension of magnetic field is

(a) $I^{-1}ML^0T^{-2}$

(b) I^0MLT^{-2}

(c) $IMLT^{-1}$

(d) $IM^{-1}L^{-1}T^{-2}$

SOL:

Correct option is A

5. Lenz's law is associated with

(a) charge

(b) mass

(c) energy

(d) principle of conservation of momentum

SOL:

Correct option is C

6. The power factor of L-R circuit is

(a) $R + WL$

(b) $\frac{R}{\sqrt{R^2 + W^2 L^2}}$

(c) $R\sqrt{R^2 + W^2 L^3}$

(d) WL / R

SOL:

Correct option is B

7. When the intensity of magnetic field is increased four times, the time-period of suspended magnetic needle becomes

(a) double

(b) half

(c) four times

(d) one-fourth less

SOL:

Correct option is B

8. The final image in astronomical telescope is

(a) real and erect

(b) real and inverted

(c) virtual and inverted

(d) virtual and erect

SOL:

Correct option is C

9. β -rays are deflected in

(a) gravitational field

(b) only in magnetic field

(c) only in electric field fields

(d) both in magnetic and electric fields

SOL:

Correct option is D

10. The Boolean expression for NOR gate is

(a) $\overline{A.B} = Y$

(b) $A + B = Y$

(c) $AB = Y$

(d) $\overline{A+B} = Y$

SOL:

Correct option is D

11. 64 identical drops each of capacity of $IMLT^{-1}$ combine to form a big what is the capacity of big drop?

(a) $25\mu F$

(b) $4\mu F$

(c) $164\mu F$

(d) $20\mu F$

SOL:

Correct option is D

12. The algebraic sum of all currents meeting at a point in an electric circuit is

(a) Zero

(b) infinite

(c) positive

(d) negative

SOL:

Correct option is A

13. Dimension of electromotive force (e.m.f) is

(a) ML^2T^{-2}

(b) $ML^2T^{-2}I^{-1}$

(c) MLT^{-2}

(d) $ML^2T^{-3}I^{-1}$

14. Ampere-hour is unit of

(a) Power

(b) Charge

(c) Energy

(d) Potential difference

SOL:

Correct option is B

15. The unit of reactance is

(a) mho

(b) ohm

(c) farad

(d) ampere

SOL:

Correct option is B

16. When a ray of light enters a glass slab, its wavelength

- (a) decreases
- (b) increases
- (c) remains unchanged
- (d) data are not complete

SOL:

Correct option is A

17. When an ammeter is shunted, its measuring range

- (a) increases
- (b) decreases
- (c) remains constant
- (d) None

SOL:

Correct option is A

18. S.I unit of self-inductance is

- (a) coulomb(C)
- (b) volt (V)
- (c) ohm (Ω)
- (d) Henry (H)

SOL:

Correct option is D

19. Hot wire ammeter measures

- (a) peak value of ac
- (b) average value of ac
- (c) r.m.s. ac
- (d) none of these

SOL:

Correct option is A

20. If N_1 and N_2 are numbers of primary and secondary coils of a step transformer, then

- (a) $N_1 > N_2$
- (b) $N_2 > N_1$

(c) $N_1 = N_2$

(d) $N_1 = 0$

SOL:

Correct option is B

21. Electron – volt (eV) is the measure of

(a) charge

(b) potential difference

(c) current

(d) electric induction.

SOL:

Correct option is D

22. The working of dynamo is based on the principle of

(a) Heating effect of current

(b) electromagnetic induction

(c) magnetic induction

(d) electric induction

SOL:

Correct option is B

23. S.I unit of magnetic moment is

(a) JT^{-2}

(b) Am^2

(c) JT

(d) Am^{-1}

SOL:

Correct option is B

24. The optical fibre works on the principle of

(a) Scattering

(b) Refraction

(c) Dispersion

(d) Total internal reflection

SOL:

Correct option is D

25. When the tube length of microscope is increased, its magnifying power

(a) increases

(b) decreases

(c) becomes zero

(d) remains unchanged.

SOL:

Correct option is A

26. The energy of a photon of wavelength λ is

(a) $hc\lambda$

(b) hc / λ

(c) $h\lambda / c$

(d) λ / hc

SOL:

Correct option is B

27. Which one of the following has maximum penetrating power?

(a) X-rays

(b) Cathode rays

(c) α -rays

(d) γ -rays

SOL:

Correct option is D

28. The majority current –carrier in p-type semiconductor is

(a) electron

(b) hole

(c) photon

(d) proton

SOL:

Correct option is B

Section-II (Non- Objective Type Question)

Question nos. 1 to 11 are of short answer type. Each question carries 2 marks. 11 x 2 = 22

1. Define dielectric strength and relative permittivity.

Sol:

Electric strength is defined as the maximum that an insulating material after loses its insulating properties. While, relative permittivity is the material property which affects

2. Explain the use of motor starter or choke coil. The Coulomb force between two point charges in the material.

Sol:

Choke Coil: A choke coil is an electrical appliance used for controlling current in an a.c circuit without wasting electrical energy in the form of heat.

Motor Starter: A motor starter is basically a variable resistor connected in series with the motor. When it is used in series with the motor, the effective resistance becomes quite large. Therefore, when the motor is switched on small current will flow even in the absence of back e.m.f.

3. Derive dimension and SI unit of magnetic flux.

Sol:

S.I unit of Magnetic flux: Tesla meter² or weber

Dimension of Magnetic flux = $[MT^2I^{-1}]$

4. State Kirchhoff's two laws for electrical network

Sol:

First law: In the network of conductor the algebraic sum of current meeting at a point is always zero. It is also called junction rule.

Second law: It states that in any closed part of an electrical circuit, the algebraic sum of the emf is equal to the algebraic sum of the products of the resistance and current flowing through them. It is also called loop rule. This law is based upon the law of conservation of energy.

5. What is total internal reflection? What are the conditions for it?

Sol:

Total Internal Reflection: The total internal reflection may be defined as the phenomenon of reflection of light that takes place, when a ray of light travelling in a denser medium gets incident at the interface of two media at an angle greater than critical angle for that pair of media.

Condition for total Internal Reflection: (i) The ray incident on the interface should be from an optically denser medium. (ii) The angle of incidence should be greater than critical angle for given pair of media.

6. A wire of resistance 16Ω is elongated to double its length. Find the new resistance.

Sol:

In this position, volume of wire remains constant

$$L_1 A_1 = L_2 A_2 = V$$

Now,

$$R_1 = \rho \frac{L_1}{A_1} = \rho \frac{L_1^2}{V} \text{-----(i)}$$

$$R_2 = \rho \frac{L_2}{A_2} = \rho \frac{L_2^2}{V} = \rho \frac{L_2^2}{V} \text{-----(ii)}$$

Dividing eqn. (ii) by (i)

$$\frac{R_2}{R_1} = \frac{\rho \frac{L_2^2}{V}}{\rho \frac{L_1^2}{V}} = \left(\frac{L_2}{L_1} \right)^2; \frac{R_2}{16} = \left(\frac{2L_1}{L_2} \right)^2 = 4; R_2 = 64 \text{ohm}$$

7. Write truth table of NAND and NOR gates.

Sol: Truth table of NAND

A	B	Y'	Y
0	0	0	1
1	0	0	1
0	1	0	1
1	1	1	0

Truth table of NOR gate

A	B	Y'	Y
0	0	0	1
1	0	1	0
0	1	1	0
1	1	1	0

8. Write Lenz's law of electromagnetic induction.

Sol:

If the changing magnetic field is linked with a coil, an emf is induced in it. This change in magnetic field may be caused by changing the magnetic field strength by moving a

magnet towards or away from the coil, or moving the coil into or out of the magnetic field as desired.

9. Write any two properties of X-rays.

Sol:

- 1) X rays are electromagnetic radiation.
- 2) Wavelength range for X ray is 0.01 to 0.1 nm.
- 3) X rays travel with the speed of light.

10. What is Peltier effect?

Sol:

Peltier effect: It is the reverse of see back effect. The absorption evolution of heat of a junction of two dissimilar metals, when a current passed, is known of peltier effect.

11. Explain sky waves and space waves.

Sol: The sky waves are the radio waves of frequency between. While, The radio waves having high frequencies are basically called as space waves

Question Nos. 12 to 15 are of long answer type. Answer must be explanatory and in your own language. All questions have alternative out of which you have to choose any one alternative.

Each question carries 5 marks.

4 x 5 =20

12. Find an expression for capacity of a parallel plate capacitor with compound dielectric.

Sol:

Let us consider a parallel plate capacitor with a dielectric media inside the plates with dielectric constant K.

Now,

σ is the surface charge density,

The electric field E

$$E = \frac{\sigma}{2K\epsilon_0} + \frac{\sigma}{2K\epsilon_0} = \frac{\sigma}{K\epsilon_0}$$

Two terms because of the two plate one will have positive and other will nejkative charge.

If d is the separation between the plates then voltage is

$$V = Ed$$

Now,

The total charge is

$$Q = \sigma A$$

Where, A is the surface area

Therefore, capacitance is

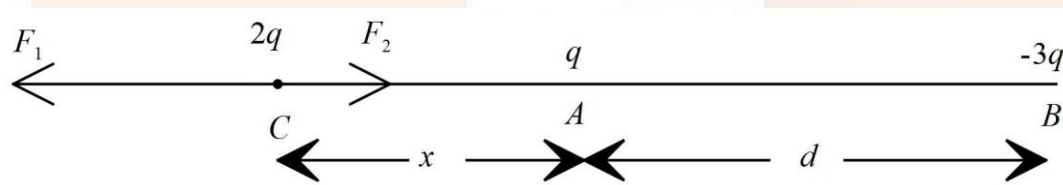
$$C = \frac{Q}{V} = \frac{\sigma A}{\frac{\sigma d}{K\epsilon_0}} = \frac{K\epsilon_0 A}{d}$$

Or

Two charges q and $-3q$ are placed fixed on X-axis at a separation d . Where should a third charge $2q$ be placed such that it will not experience any force? Also calculate potential energy of the system.

Sol:

Let third charge $2q$ is placed at the x from charge q in left side. In the position force exerted by charge q on $2q$ is equal in magnitude but opposite in direction of force exerted by charge $-3q$.



$$F_1 = F_2; \frac{1}{4\pi\epsilon_0} \cdot \frac{q \cdot 2q}{x^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{2q \cdot 3q}{(x+d)^2}; \frac{1}{x^2} = \frac{3}{(x+d)^2}$$

$$\frac{1}{x} = \frac{\sqrt{3}}{x+d}; \sqrt{3}x = x + d \sqrt{3}x - x = d;$$

$$x = \frac{d}{(\sqrt{3}-1)}$$

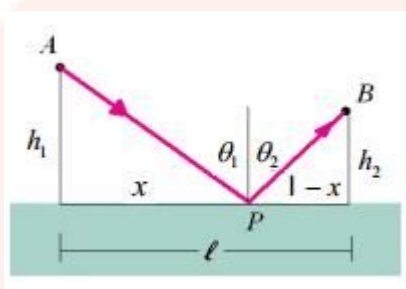
Charge is placed at the distance $\frac{d}{(\sqrt{3}-1)}$ metre.

13. How are wave front and secondary wavelets defined? Verify laws of reflection or laws of refraction on the basis of Huygens' wave theory.

Sol:

Consider the light ray shown in the figure. A ray of light starting at point A reflects off the surface at point P before arriving at point B, a horizontal distance from point A. we calculate the length of each path and divide the length by the speed of light to determine the time required for the light to travel between the two points.

$$t = \frac{\sqrt{x^2 + h_1^2}}{c} + \frac{\sqrt{(1-x)^2 + h_2^2}}{c}$$



To minimize the time we set the derivative of the time with respect to x equal to zero. We also use the definition of the sine as opposite side over hypotenous to relate the lengths to the angles of incident and reflection.

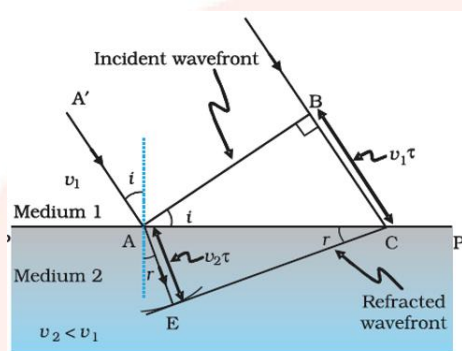
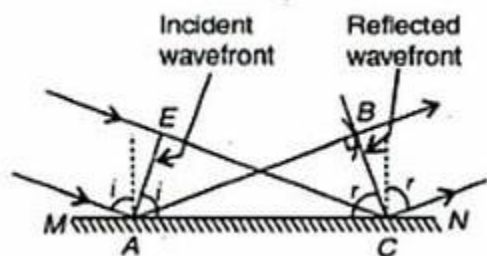
$$0 = \frac{dt}{dx} = \frac{x}{\sqrt{x^2 + h_1^2}c} + \frac{(1-x)}{\sqrt{(1-x)^2 + h_2^2}c}$$

$$\frac{x}{\sqrt{x^2 + h_1^2}c} = \frac{(1-x)}{\sqrt{(1-x)^2 + h_2^2}c} \rightarrow \sin \theta_1 = \sin \theta_2 \rightarrow \theta_1 = \theta_2$$

Consider a plane wave AB incident at an angle i on a reflecting surface MN. If v is the speed of the wave in the medium and t is the time taken by the wavefront to cover the distance BC, then $BC = vt$

To construct the reflected wavefront we draw an arc (represent reflected wavefront) of radius vt from the point A.

Draw a tangent on the arc . we obtain $AE = BC = vt$



We can see a ray of light is incident on this surface and another ray which is parallel to this ray is also incident on this surface. As these rays are incident from the surface, so we call it incident ray.

Let PP' represent the medium 1 and medium 2. The speed of the light in this medium is represented by v_1 and v_2 . If we draw a perpendicular from point 'A' to this ray of light, Point A, and point B will have a line joining them and this is called as wavefront and this wavefront is incident on the surface.

If 'r' represents the time taken by the wavefront from the point B to C then the distance,

So to determine the shape of the refracted wavefront, we draw a sphere of radius $v_2 r$ from the point A in the second medium. Let CE represent a tangent plane drawn from the point C on to the sphere. Then, and CE would represent the refracted wavefront. If we now consider the

triangles ABC and AEC, we get that,

$$\sin i = \frac{BC}{AC} = \frac{v_1 r}{AC}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2 r}{AC}$$

Where, i and r are the angles of incidence and refraction

$$n_1 \sin i = n_2 \sin r$$

Or,

What is photoelectric emission effect? What are the laws of photoelectric emission effect? How did Einstein explain these laws of photoelectric effect?

Sol:

$m = \frac{v}{u}$ Here Real and virtual image are possible

$v = +3u$ and $-3u$ (for virtual and Real image respectively)

By lens formula for virtual image

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}; \frac{1}{3u} - \frac{1}{-u} = \frac{1}{15}; \frac{-1+3}{3u} = \frac{1}{15}; \frac{+2}{3u} = \frac{1}{15}; 4 = 10\text{cm}$$

For Real image $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}; \frac{1}{3u} - \frac{1}{-u} = -\frac{1}{15}; \frac{4}{3u} = \frac{1}{15}; 4 = 20\text{cm}$

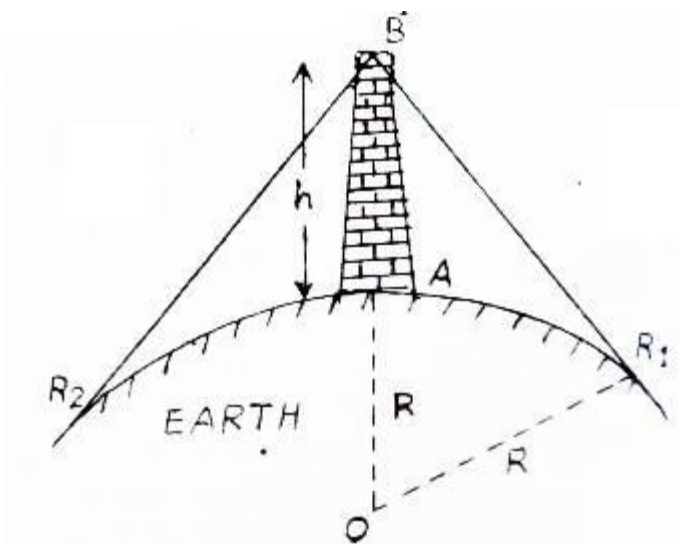
Therefore, object is placed of distance 20 cm for real image and at 10cm for virtual image.

14. Describe amplitude modulation and frequency modulation. Derive an expression for covering range of transmission antenna.

Sol:

Amplitude modulation: When the modulating wave is superimposed on a high frequency carrier wave in a manner that the frequency of modulated wave is same as that of the carrier wave but its amplitude is modified in accordance with that of the modulating wave, the process is called amplitude modulation.

Frequency modulation: When the modulating wave is superimposed on a high frequency carrier wave in a manner that the amplitude of modulated wave is same as that of the carrier wave but its frequency is modified in accordance with the amplitude of the modulating wave, the process is called frequency modulation.



Expression for height of Antenna: Let h is the height of Antenna AB . Due to curvature of the earth, the transmitted signal can be received up to maximum distance AR_1 (or AR_2) from the antenna.

Let $AR_1 = AR_2 = d$

Therefore, the TV signal will be received in a circuit of radius d . let us derive the relation between height (h) of TV Antenna and maximum distance (d) up to which the TV signal can be received. From right angles $\triangle OBR_1$

$$(OB)^2 = (BR_1)^2 + (OR_1)^2$$

Taking $BR_1 = AR_1 = d, (R+h)^2 = d^2 + R^2; d^2 = h^2 + 2hR$

Since, h is small as compared to the radius of the earth, h^2 can be neglected as compared to $2hR$. Therefore,

$$d^2 = 2hR$$

$$d = \sqrt{2hR}$$

This is the expression for height of Antenna.

OR

What is photoelectric emission effect? What are the laws of photoelectric emission effect? How did Einstein explain these laws of photoelectric effect?

Sol:

Photoelectric effect is the phenomenon of emission of electrons from the surfaces of certain substances, mainly metals, when light of shorter wavelength is incident upon them.

The four laws of photoelectric emission are given as:

LAW 1: For a given metal and frequency, the number of photoelectrons emitted is directly proportional to the intensity of the incident radiation.

LAW 2: For a given metal, there exists a certain frequency below which the photoelectric emission does not take place. This frequency is called threshold frequency.

LAW 3: For a frequency greater than the threshold frequency, the kinetic energy of photoelectrons is dependent upon frequency or wavelength but not on the intensity of light.

LAW 4: Photoelectric emission is an instantaneous process. The time lag between incidence of radiations and emission of electron is 10^{-9} seconds.

15. Define self-inductance and mutual inductance. Find an expression for mutual inductance of two long coaxial solenoids.

Sol: **Self-inductance or co-efficient of self-induction:** It is found that magnetic flux ϕ linked with coil is directly proportional to current passing through it.

$$\phi \propto I; \phi = LI$$

Where L is constant and it is called co-efficient of self-induction. It

$$I = 1 \text{ Then } \phi = L$$

Thus self-inductance of a coil is numerically equal to the magnetic flux linked with the coil when a unit current flows through it.

Mutual Inductance or co-efficient of mutual induction: It is found that the magnetic flux ϕ linked with S-coil at that instant. Then

$$\phi_s \propto I_p; \phi_s = MI_p$$

Where M is constant and it is called co-efficient of mutual induction or mutual inductance.

$$\text{It } I_p = 1 \text{ then } \phi_s = M$$

Thus, mutual inductance of two coil is numerically equal to magnetic flux linked with one coil, when a unit current flows through the neighboring coil.

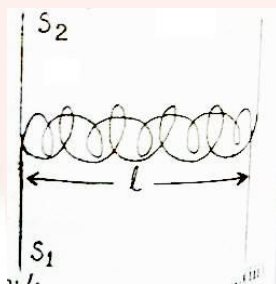
Expression for mutual inductance of two coil: Consider two long solenoid s_1 and s_2 of same length l such that the solenoid s_1 Completely. Let n_1 and n_2 be the number of turn per unit length of solenoid s_1 and s_2 . Current I_1 is passed through the solenoid S_1 . Then due to flow of current through s_1 , magnetic fields will be produced inside it and magnetic flux will linked with coil s_2 . The magnetic flux(ϕ_{21})linked with solenoid s_2 is directly proportional to current I_1 .

$$\phi_{21} \propto I_1; \quad \phi_{21} = M_{21} I_1 \text{ -----(i)}$$

Where M_{21} is co-efficient of mutual induction of the two solenoid.

Now, magnetic field produced inside the solenoid S_1 on passing current through it.

$$B_1 = \mu_0 n_1 I_1 \text{ -----(ii)}$$



The magnetic flux linked with each turn of the solenoid $S_2 = B_1 A$

Therefore total magnetic flux linked with solenoid S_2

$$\begin{aligned} \phi_{21} &= B_1 A \times n_2 l = \mu_0 n_1 l \times n_2 l \text{ -----(iii)} \\ \phi_{21} &= \mu_0 n_1 n_2 A l I_1 \end{aligned}$$

From eqn. (i) and (iii)

$$M_{21} = \mu_0 n_1 n_2 A l \text{ -----(iv)}$$

Similarly mutual inductance from coil S_2

$$M_{12} = \mu_0 n_1 n_2 A l \text{ -----(v)}$$

From eqn. (iv) and (v) it follows that in case of two long solenoids, the co-efficient of mutual induction remains same.

Therefore, $M_{21} = M_{12} = M$

Hence, co-efficient of mutual induction between two long solenoid

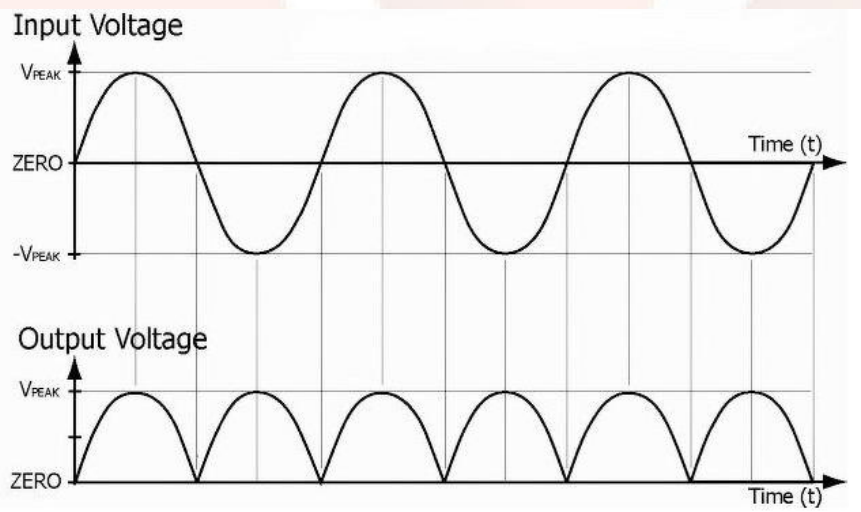
$$M = \mu_0 n_1 n_2 A l$$

OR

What is rectifier? How a p-n junction diode can be used in (i) forward biased and (ii) reverse biased mode? Draw a labeled $V - I$ characteristic of p-n junction diode on a graph.

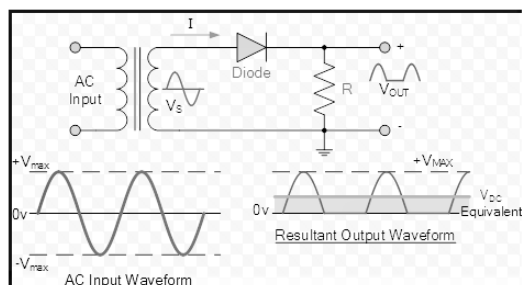
Sol:

Rectifier: An electronic device which converts a.c power in to d.c. power is called Rectifier.



Forward bias: when an external d.c source is connected to the diode with p-section connected to positive pole and n-section to the negative pole, the junction diode is said to be forward bias.

It makes the depletion layer thin and as then, the junction diode offers low resistance during forward bias.



Reverse bias: when a battery is connected to junction diode with p-section connected to negative pole and n-section connected to positive pole, the junction diode is said to be reverse biased.

It makes the depletion layer thick and the junction diode offers high resistance during reverse bias.

V-I Characteristic of p-n junction

